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	This project utilizes the genetic model organism, Drosophila melanogaster, the fruit fly, to explore the response of multicellular organisms to extreme environmental change and to identify factors that determine or modify the ability of this model system to tolerate environmental change. This work has focused primarily on physiological pathways that are known to be responsive to stressful conditions in mammals, pathways which utilize the neurotransmitter and neurohormonal family of chemicals, the catecholamines. These are highly conserved pathways. Very similar genes and protein products of these genes are found in all organisms with nervous systems. The project has been based on three hypotheses. The first was that catecholamine production is regulated in response to environmental stress in Drosophila as it is in mammals – an essential criterion if Drosophila is to be a useful model. The second was that regulation of the catecholamine pathway is critical to an organism's ability to tolerate stressful environments. This hypothesis is one that		

Task Description:	the Drosophila system is uniquely suited to address since a large collection of strains with mutations in the critical genes makes possible the testing of physiological consequences for stress tolerance in strains with perturbed catecholamine regulation. The third hypothesis was that gravitational change, even though it is not an environmental condition normally encountered by Drosophila, would trigger responses that are similar or identical to those caused by a variety of stressors that are encountered in its normal environment. This project has three specific goals: (1) To determine whether mutation-induced changes in catecholamine regulation affect ability to tolerate environmental change, aging, behavior and fertility. (2) To determine the molecular genetic responses of catecholamine-regulating genes to environmental changes. (3) To identify cell types and developmental stages most responsive to and affected by environmental change. Space travel subjects humans and other organisms to extreme environmental change. It is critical to understand cellular mechanisms and genetic variation that can dictate the tolerance of organisms to such change. We have identified genes in Drosophila whose normal function and regulation is crucial for the ability to cope with stress, and these genes are virtually identical in sequence and function to their human counterparts. Moreover, we have found that dietary supplementation with the products of the pathways in question can modify the ability to withstand environmental change. Use of Drosophila as a model system for this study allows the application of a powerful array of genetic and molecular technologies, which in this instance, have provided insights into genetic variation that directly modifies the ability of this organism to cope with environmental extremes. The high degree of genetic conservation among multicellular organisms means that it is very likely that our results can be extrapolated to higher organisms. The wealth of genetic materials, the ease of rearing vast numbers of organisms in very small spaces, and the low cost of producing and maintaining mutant strains also contribute strong arguments for pursuing these studies in this model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A key feature of the ability of animals to convert perception of environmental change into appropriate metabolic responses is the system that regulates the synthesis of catecholamine neurotransmitters and neurohormones. These responses allow organisms to adjust to new environments, to make protective physiological changes when harmful environments are encountered, and to initiate appropriate behavioral changes. Regulation of the production of catecholamines is particularly evident under conditions of extreme environmental variation. While these metabolic responses allow organisms to tolerate quite disparate environmental conditions in many instances, they can also have negative consequences on health and reproduction. Such consequences are due to the diverse signaling functions that catecholamines have in all animal species. In vertebrates, they have been implicated as signaling molecules in cardiac development and in both vertebrates and insects, roles in ovarian development have been observed. Perturbations in catecholamine levels cause abnormalities in sensory development and function, in neuromuscular and motor neuron activity and in certain reproductive behaviors in Drosophila. It is therefore crucial to identify the molecular components of the catecholamine pathway that participate in regulating responses to environmental change and to ascertain the physiological consequences of the catecholamine response in more detail. Our work this year has lead to several striking and unexpected results that have important implications with respect to the long term effects of stress because some of the changes that we observe in our mutants, changes that enhanced by stress, are molecular changes that may be implicated in addictive behaviors and others in the progress of Parkinson"s Disease. We anticipate that these studies will lead to strong mechanistic descriptions of how this organism responds and adjusts to environmental conditions. We expect to better understand the role that the catecholamine pathway plays in physiological alterations associated with flight conditions. Because this pathway and the genes involved are highly conserved, it is expected that this study will provide more generally applicable insights into all animal species' responses to environmental change.
Task Progress:	Studies in other laboratories demonstrated that hypergravity conditions affect aging and stress resistance in Drosophila. We have shown that the pathway regulating the catecholamine, dopamine, is one of the cellular systems implicated in these effects. Our work during this year has utilized various mutant strains that have altered dopamine regulation. These genes are (1) Punch, which encodes GTP cyclohydrolase, an enzyme required for the production of a cofactor, tetrahydrobiopterin (BH4) that is required for and positively regulates catecholamine expression. (2) pale, which encodes tyrosine hydroxylase, the first enzyme in the pathway that synthesizes dopamine. It is the rate limiting component in the pathway, but is directly regulated by cofactor availability. (3)Catsup, which encodes a membrane protein called Catecholamines-up; this protein acts as a negative regulator of GTP cyclohydrolase and tyrosine hydroxylase. We have also utilized wild type strains to determine the normal biochemical responses to environmental change. Key experiments and results are summarized below: I. We have completed experiments to characterize the effect of environmental change on dopamine and BH4 levels. We find that both products are strongly induced by starvation, one of the stressors employed in our studies. During the course of the first 24 hrs of food deprivation, these levels increase. By 48 hrs, DA levels drop, associated with the loss of dopamine –containing neurons in the central nervous system. We have initiated a detailed study of these compounds during exposure to elevated gravity. We have found that levels change. As in other environmental stresses, the responses are highly dynamic, and we will be adding additional times of exposure in order to clarify the pattern of responses. Nevertheless, our data allow the conclusion that in Drosophila, DA and BH4 synthesis are extremely responsive to environmental change. We then went on to assay GTP cyclohydrolase and tyrosine hydroxylase, the rate-limiting enzymes for BH4 and dopamine synthesis, respectively. We observe changes in the activities of the enzymes corresponding to the changes in metabolite levels. II. Oxidative stress has been implicated as a causative agent in various neurodegenerative diseases and in aging. Previously, we determined that mutant strains in which dopamine levels are reduced are more sensitive to environmental stress, while our strains that have elevated levels of dopamine are very resistant to stressors that include starvation, dessication, heat and hypergravity. Thus, we not only observe an organismal response to these stressors in which both BH4 and DA are elevated, but we can associate them with a protective function, since the consequences of pathway perturbation directly correlate with stress tolerance. In addition, we have found a correspondence with respect to life span. Catecholamines up mutants, which have elevated BH4 and DA pools, live significantly longer than wild type controls, while dopamine deficient strains have reduced life span. We therefore directly addressed whether these mutants exhibited modification in their response to oxidative stress. We did this by allowing mutant and wild type control individuals to ingest low doses of paraquat, an agent know to increase the generation of reactive oxygen species in mammalian cells. Our studies clearly demonstrated that mutants with elevated dopamine production are significantly more resistant to paraquat exposure while low dopamine mutants are highly sensitive to this treatment. Because dopamine that remains within a cell can, itself, be converted to reactive oxygen species, this result initially appeared

counterintuitive. However, since behavioral studies demonstrate that high dopamine levels in these mutant strains are associated with behavioral changes, including hyperactivity, we infer that the excess dopamine is actively released from dopaminergic cells and is thus likely to be compartmentalized, biologically active and inaccessible for the generation of reactive species.

III. We then asked whether ingestion of either dopamine or BH4 could alter stress responses. We find that both can provide significant protection against stress. Ingestion of either compound strongly enhances tolerance to starvation and to oxidative stress. Thus, we have clearcut evidence that the adverse effects of extreme environments can be ameliorated to a significant extent by simply feeding the organism either of these products. We have now turned to an examination of the effects of DA on responses to and effects of exposure to hypergravity conditions. These studies have been completed and the data are being analyzed at this time. We have extended pharmacological approaches to the modification of stress protection by feeding wild type Drosophila an inhibitor of tyrosine hydroxylase activity, 3-iodotyrosine, in sublethal concentrations. In accordance with the results of our stress tolerance assays of mutant strains with dopamine defects, we also observe that inhibition of tyrosine hydroxylase activity with 3-iodotyrosine also results in increased stress sensitivity.

IV. Most recently, we have begun exploring the possible induction of free radical scavenging enzymes in Drosophila brain tissue during extended periods in environmental extremes. We are beginning to assess whether dopamine perturbations in the mutant strains are associated with changes in scavenging enzyme activity in normal and stressful environments.

V. We have been attempting to construct transgenic strains that use Green Fluorescent Protein expression as a reporter for stress response, employing a GTP cyclohydrolase gene promoter to drive expression. Transgenic lines with about 1 kilobase of upstream sequence fail to express this protein at significant levels. We are currently restructuring the reporter construct adding additional sequences upstream of the transcription unit. This gene is highly regulated, with extensive temporal and spatial specificity and dynamic expression patterns. Thus, we anticipate that accurate expression will require significant upstream sequence. However, once construction of strains is achieved, these lines will allow an in vivo assay of response to environmental change.

Bibliography Type:	Description: (Last Updated: 08/06/2008)
Books/Book Chapters	O’Donnell, J. M., Wang, Zand Chaudhuri, A. "Effects of perturbation of catecholamine regulation on resistance of Drosophila melanogaster to environmental stress." in "Pterins, Folates, and Related Biogenic Amines " Ed. Blau, N and Beat T. SPS Publications, 94-100 , Dec-2003